843: HELMUTKOENIGIA MEDIUM

NaCl	1.00	g
KCl	0.50	g
$MgCl_2$ x 6 H_2O	0.40	g
$CaCl_2$ x 2 H_2O	0.10	g
NH_4Cl	0.30	g
KH_2PO_4	0.20	g
Na_2SO_4	0.15	g
Trace element solution SL-10	1.00	ml
Selenite-tungstate solution	1.00	ml
Trypticase peptone (BD BBL)	1.00	g
Sodium resazurin (0.1% w/v)	0.50	ml
Na_2CO_3	1.50	g
Sugar mix	30.00	ml
Yeast autolysate	50.00	ml
Seven vitamins solution	1.00	ml
Co-factors solution	20.00	ml
DL-Dithiothreitol (DTT)	0.40	g
Distilled water	900.00	ml

1. Dissolve ingredients (except carbonate, sugar mix, yeast autolysate, vitamins, co-factors, and DTT), sparge medium with 80% N_2 and 20% CO_2 gas mixture for 30 - 45 min to make it anoxic. Distribute under same gas atmosphere into anoxic Hungate-type tubes or serum vials to 30% of their volume (e.g., 4.5 ml per Hungate-type tube) and autoclave. After sterilization add carbonate from a sterile anoxic stock solution prepared under 80% N_2 and 20% CO_2 gas atmosphere and sugar mix, yeast autolysate, vitamins, co-factors and DTT from sterile anoxic stock solutions prepared under 100% N_2 gas atmosphere and sterilized by filtration. Check pH of complete medium and adjust to 7.2 - 7.4, if necessary.

2. Note: Basal medium and yeast autolysate have to be prepared freshly and cannot be stored for longer periods.

Sugar mix (from medium 843)

Cellobiose	56.70	g
Sucrose	56.70	g
Maltose x H_2O	60.00	g
Trehalose	63.30	g
D-Xylose	25.00	g
Distilled water	1000.00	ml

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Yeast autolysate

Yeast extract (dry baker's yeast, SIGMA YSC-1)	280.00	g
Distilled water	1000.00	ml

Suspend dry baker's yeast (SIGMA YSC-1) in distilled water and allow cells to autolyse for 48 h at 55°C. Thereafter, cell debris are removed by centrifugation (3.700 x g for 30 min) and the amber-colored supernatant is stored in aliquots at -20°C.

Co-factors solution

Thiamine pyrophosphate	2.50	g
Pyridoxal hydrochloride	0.25	g
Pyridoxal phosphate	0.25	g
Ca-folinate	0.05	g
ß-NAD	0.05	g
Coenzyme A	0.05	g
FAD	0.05	g
Haemin (dissolved separately in 10 mM NaOH)	65.00	mg
Nicotinamide	25.00	mg
Folic acid	2.50	mg
Riboflavin	0.50	mg
Distilled water	1000.00	ml

Adjust pH to 7.0 with 1 N NaOH against indicator paper, filter sterilize and store under 100% N_2 gas atmosphere.

Trace element solution SL-10 (from medium 320)

HCl (25%)	10.00	ml
$FeCl_2 \times 4\ H_2O$	1.50	g
$ZnCl_2$	70.00	mg
$MnCl_2 \times 4\ H_2O$	100.00	mg
H_3BO_3	6.00	mg
$CoCl_2 \times 6\ H_2O$	190.00	mg
$CuCl_2 \times 2\ H_2O$	2.00	mg
$NiCl_2 \times 6\ H_2O$	24.00	mg
$Na_2MoO_4 \times 2\ H_2O$	36.00	mg
Distilled water	990.00	ml

First dissolve $FeCl_2$ in the HCl, then dilute in water, add and dissolve the other salts. Finally make up to 1000.00 ml.

Selenite-tungstate solution (from medium 385)

NaOH	0.50	g

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Na_2SeO_3 x 5 H_2O	3.00	mg
Na_2WO_4 x 2 H_2O	4.00	mg
Distilled water	1000.00	ml

Seven vitamins solution (from medium 503)

Vitamin B_{12}	100.00	mg
p-Aminobenzoic acid	80.00	mg
D-(+)-biotin	20.00	mg
Nicotinic acid	200.00	mg
Calcium pantothenate	100.00	mg
Pyridoxine hydrochloride	300.00	mg
Thiamine-HCl x 2 H_2O	200.00	mg
Distilled water	1000.00	ml

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